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Deficiency: Die back of citrus, Reclamation disease of cereals and legumes, chlorosis, necrosis and Exanthema in *Citrus*.

4. **Zinc (Zn):** Essential for the synthesis of Indole acetic acid (Auxin) activator of carboxylases, alcohol dehydrogenase, lactic dehydrogenase, glutamic acid dehydrogenase, carboxy peptidases and tryptophan synthetase. It is absorbed as Zn^{2+} ions.

   Deficiency: Little leaf and mottle leaf due to deficiency of auxin, Inter veinal chlorosis, stunted growth, necrosis and Khaira disease of rice.

5. **Boron (B):** Translocation of carbohydrates, uptake and utilisation of Ca^{++}, pollen germination, nitrogen metabolism, fat metabolism, cell elongation and differentiation. It is absorbed as borate BO^{3-} ions.

   Deficiency: Death of root and shoot tips, premature fall of flowers and fruits, brown heart of beet root, internal cork of apple and fruit cracks.

6. **Molybdenum (Mo):** Component of nitrogenase, nitrate reductase, involved in nitrogen metabolism, and nitrogen fixation. It is absorbed as molybdate (Mo^{2+}) ions.

   Deficiency: Chlorosis, necrosis, delayed flowering, retarded growth and whip tail disease of cauliflower.

7. **Chlorine (Cl):** It is involved in Anion – Cation balance, cell division, photolysis of water. It is absorbed as Cl^- ions.

   Deficiency: Wilting of leaf tips

8. **Nickel (Ni):** Cofactor for enzyme urease and hydrogenase.

   Deficiency: Necrosis of leaf tips.



Calmodulin

Calmodulin is a Ca^{2+} modulating protein in eukaryotic cells. It is a heat stable protein involved in fine metabolic regulations.

12.4 Deficiency diseases and symptoms

The following table (Table 12.2) gives you an idea about Minerals and their Deficiency symptoms:

Activity

Collect leaves showing mineral deficiency. Tabulate the symptoms like Marginal Chlorosis, Interveinal Chlorosis, Necrotic leaves, Anthocyanin formation in leaf, Little leaf and Hooked leaf. (Discuss with your teacher about the deficiency of minerals)

12.5 Critical concentration and toxicity of minerals

12.5.1 Critical Concentration

To increase the productivity and also to avoid mineral toxicity knowledge of critical concentration is essential. Mineral nutrients lesser than critical concentration cause deficiency symptoms. Increase of mineral nutrients more than the normal concentration causes toxicity. A concentration, at which 10 % of the dry weight of tissue is reduced, is considered as toxic. Figure 12.2 explains about Critical Concentration.